

FREDRICK BEMDOO JOHN

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AI and Machine learning focused Computer Engineering undergraduate with over 3 years of active engagement in technology and 2+ years of involvement in AI, data analysis and research. International engineering remote internship experience with ITU-T in Switzerland, co-authorship of a PSO-ANFIS precision agriculture paper submitted to IEMIS 2026, award-winning hackathon experience, and a track record of leading teams and training 300+ students Hands-on experience across AI model development, MATLAB/Simulink, Python, embedded systems, data analytics, technical research, and production-oriented software development.

EXPERIENCE

JULY 2026 – SEPTEMBER 2026

Engineering Intern | ITU-T, TSB Operations & Planning, Switzerland

Activities: Built an end-to-end AI pipeline to extract member-organisation declarations from unstructured ITU working documents and resolve official ITAD organisation IDs.

APRIL 2023 – APRIL 2023

SOCIAL MEDIA CONSULTANT, HEALTHY TREAT'S CONFECTIONARY, BWARI KITCHEN, BWARI ABUJA.

Activities: Guided and provided strategies on how to build stronger online presence, engagement and brand awareness.

SEPTEMBER 2023 – TILL DATE

RESEARCH ASSISTANT IN MINNA, GWIN RESEARCH GROUP MINNA NIGERIA.

Activities: Paper and technical writings.

SEPTEMBER 2020 – FEBRUARY 2023

FORM TEACHER, De-Gopherwood Montessori School Kuduru Road Bwari, Abuja.

Activities: Taught and oversaw the academic progress of pupils.

NOVEMBER 2025 – DECEMBER 2025

AI/ML Project Developer | Independent / Academic Projects

Activities: Design and develop AI/ML systems spanning agriculture, personal safety, decision support, computer vision, chatbots, and engineering applications

EDUCATION

MAY 2023- TILL DATE

B.ENG COMPUTER ENGINEERING (IN-VIEW), FEDERAL UNIVERSITY OF TECHNOLOGY MINNA, NIGER STATE.

SEPTEMBER 2017- AUGUST 2020

WASSCE/SENIOR SCHOOL CERTIFICATE EXAMINATION, GOVERNMENT SECONDARY SCHOOL BWARI, ABUJA.

SEPTEMBER 2014- JUNE 2017

WASSCE/JUNIOR SCHOOL CERTIFICATE EXAMINATION, JUNIOR SECONDARY SCHOOL BWARI 1, ABUJA.

SEPTEMBER 2008- APRIL 2014

FIRST SCHOOL LEAVING CERTIFICATE, MODEL PRIMARY SCHOOL BWARI, ABUJA.

CORE TECHNICAL SKILLS

- Artificial Intelligence & Machine Learning: supervised/Unsupervised ML, Deep Learning, BI-LSTM. Fuzzy Logic systems
- Expertise in leveraging Generative AI (ChatGPT, Claude, Gemini, Kimi, Lovable) to automate workflows.
- Effective communicator with excellent planning, organizational, and ability to lead, motivate, inspire excellence, establish goals, and attain result
- Passionate for learning and leveraging on opportunities to expand my knowledge and skillset.

- Programming & Tools: Python, MATLAB, Simulink, SQL, C, C++, Arduino.
- AI Development: Model training, inference pipelines, chatbot development, AI-powered applications.
- Data & BI Tools: Tableau, Power BI, Python (Pandas, NumPy, Matplotlib)
- Proficient in n8n, Make.com and Pabbly connect for building complex, scalable automations, API integrations and multi-step workflows
- Technical proficiency in research tools, E-draw, Visio, Mendeley reference manager, etc.
- Technical proficiency in the use of: web development tools, HTML CSS, web development templates like WIX in building websites
- Collaboration & leadership: Team leadership, project coordination, mentoring, technical training.

RESEARCH & PUBLICATION

- Co-author, “Smart Multiseed Precision Planting Controller” — PSO-ANFIS research project submitted to IEMIS 2026 for Springer publication.
- Designed and trained an Adaptive Neuro-Fuzzy Inference System (ANFIS) in MATLAB to determine seed spacing and seed count across groundnut, maize, and wheat based on crop and soil type.
- Optimized the ANFIS model using Particle Swarm Optimization (PSO) and integrated the controller into an Arduino Mega-based robotic planter architecture.
- Project spans applied AI/fuzzy logic, optimization, embedded systems, hardware design, statistical validation, and academic technical writing.

CERTIFICATIONS

- Certificate in Artificial Intelligence Basics – Digital Skills Nigeria
- Certificate in Artificial Intelligence & Machine Learning - Machine Learning Onramp
- Certificate in Deep Learning - Deep Learning Onramp
- Certificate in MATLAB - MATLAB Onramp
- Certificate of Internship completion ITU-T

KEY TECHNICAL PROJECTS

- **AI-Powered Personal Safety Application (Hackathon Project)**
Built a proactive safety app that listens to environmental speech, converts audio to text, processes context using a Bi-LSTM model, detects threats, and triggers automatic alarms with emergency notifications to saved contacts.
- **AI Crop Planner MATLAB Application**
Developed an intelligent crop-planning system using AI models to recommend optimal planting strategies based on data inputs.
- **AI Telegram Chatbots**
Designed and deployed intelligent Telegram bots using Python and Telegram API for automated decision support and recommendations.
- **Smart Microgrid Simulation (Fuzzy Logic System)**
Designed a fuzzy logic-based smart microgrid in MATLAB/Simulink, optimizing energy distribution and stability for renewable integration.
- **DEPSO Optimization for UPFC Placement (MATLAB Simulations)**
Developed MATLAB simulations using DEPSO to optimally place and size UPFCs, reducing congestion costs in a rescheduled generator transmission network.
- **AI Face Recognition Device | AI / Embedded Systems |**

Built an AI-based face recognition device integrating machine learning with hardware components.

COMPETITIONS & HACKATHONS

- 3rd Place Winner, AI Hackathon (Regional Level) – 2025
Sponsored by Deep Funding.
- Participant, MATLAB Coding Challenge 2025.
- Participant, SAND AI Challenge 2025 – Worked on early detection of ALS disease.
- Participant, MCP Competition 2025 - Worked on a Tap-to-Pay AI micro-credit system.

LEADERSHIP & COMMUNITY EXPERIENCE

- Community Lead, FUTMINNA MATLAB Community
- MATLAB AI & Machine Learning Team Lead
- Head, Research & Development Team, Computer Engineering
- Team Lead, SOLUTION MATRIX — MATLAB Coding Challenge 2024
- Coordinator, MATLAB Interns FUTMINNA 2024
- Financial Secretary, NACOMES FUTMINNA — improved financial coordination and accountability.

- Course Representative, Computer Engineering
- Organiser, CPE Digital Skill Set — coordinates student digital-skills acquisition programmes.
- Volunteer, NITDA Hub FUTMINNA — supports MATLAB/Simulink training and student activities.
- Volunteer, GWiN Research Group — research and technical writing.
- Trained and mentored 300+ students in MATLAB, web development, research writing, and technology-related skills

ACTIVITIES

APRIL 2024- PRESENT

VOLUNTEER, NITDA HUB, FEDERAL UNIVERSITY OF TECHNOLOGY, MINNA.

Activities: ASSISTING IN ORGANISING ACTIVITIES AND TRAINING OF STUDENTS ON THE USE OF MATLAB AND SIMULINK.

JANUARY 2024- PRESENT

VOLUNTEER, Computer Engineering (CPE) DIGITAL SKILL SET

Activities: Organizing digital skill acquisition programs for students.

FEBRUARY 2022- AUGUST 2025

VOLUNTEER, CPE TUTORIALS

Activities: Organizing academic tutorials for students.

MARCH 2022- PRESENT

VOLUNTEER, GWiN RESEARCH GROUP.

Activities: Paper and technical writings.

PERSONAL DATA

Birth: 3rd August 2003

Place: Bwari, Abuja.

Nationality: Nigerian
Sex: Male
Marital Status: Single
Languages: English.

REFEREES

Available on request